SALSABIL OUNI

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PROFESSIONAL EXPERIENCE

Farhat Hached Hospital — **2026/2 - Present**
Structural Genomics and Protein Bioinformatics Researcher — **Sousse, Tunisia**

- Built a pipeline linking experimental **ΔΔG** data to **FoldX** alanine-scan hotspot patches on **p53**, showing destabilizing variants are enriched in structural microenvironments and replicating this across independent datasets.
- Ensured rigor via confounder-adjusted regression, distance/threshold-sensitivity analyses, and cross-assay replication; validated by recovering **p53**'s known functional patches in **PyMOL**.

Arcadia — **2025/7 - 2025/9**
Machine Learning Engineer Intern — **Osaka, Japan**

- Developed an Arabic Text-to-Speech system using **VITS** (Variational Inference with adversarial learning), combining Conditional **VAE** and **GAN**-based architectures to generate natural speech with human-like prosody achieving **97%** intelligibility based on Spectrogram analysis
- Deployed a real-time speech synthesis platform using **Python, Svelte, and Docker** with **<1.5s** latency.

PURA Solutions — **2024/6 - 2024/10**
Artificial Intelligence Engineer Intern — **Sousse, Tunisia**

- Built a medical AI system for **Thoracic X-ray** analysis and report generation using **CNN** and **Transformer** architectures, achieving **92%** validation performance.
- Developed an interactive deployment pipeline using Streamlit for model evaluation and visualization.

Systems Nakashima — **2024/2 - 2024/5**
Machine Learning and Full Stack Research Engineer Intern — **Tokyo, Japan**

- Developed **Genki Walk**, an AI rehabilitation system used by **36+ patients**, combining **MediaPipe pose estimation and Transformer models** for gait analysis.
- Built and deployed a healthcare platform using **Django, Vue.js, and AWS**, achieving **91% gait analysis accuracy**.

Systems Nakashima — **2023/8 - 2023/10**
Artificial Intelligence and Full Stack Engineer Intern — **Tokyo, Japan**

- Developed **Aintone**, an AI-powered sales forecasting platform using **PyTorch** based machine learning **models** for predictive analytics and built an end-to-end application integrating **Django** backend and **Vue.js** frontend.
- Presented the project at **Tokyo Big Sight 2023**, attracting **400+** attendees and receiving positive feedback from industry professionals.

EDUCATION

Faculty of Medicine Ibn El Jazzar of Sousse — **2024 - Present**
Master's Degree in Molecular Medicine : AI Applied to Genetics, Immunology and Microbiology

Polytechnic School of Sousse — **2024 - Present**
Engineering Diploma in Artificial Intelligence and Data Science

Higher Institute of Applied Science and Technology of Sousse — **2021 - 2024**
Bachelor of Computer Science and Software Engineering,

ADDITIONAL INFORMATION

Technical Skills: Python | PyTorch | Transformers | Hugging Face | LLMs | RAG | Machine Learning | Deep Learning | NLP | Computer Vision | Scikit-learn | NumPy | Pandas | SciPy | Biopython | Computational Genomics | Protein Bioinformatics | AlphaFold | FoldX | PyMOL | Modeller | SQL | PostgreSQL | Docker | Kubernetes | Git | AWS | Linux | Bash | Spark | GitHub Actions | GitLab CI | R | C/C++ | Java

Languages: English | French | Arabic | Japanese | German

Certifications: NVIDIA Fundamentals of Deep Learning | NVIDIA Introduction to Transformer-Based Natural Language Processing | NVIDIA Generative AI with Diffusion Models | Microsoft Azure Data Fundamentals

Achievements: AIESEC Waseda Japan Global Talent 2023, selected out of 22 countries globally | AI Project Exhibitor at Tokyo Big Sight 2023 | Winner of 16 Hackathons.

Projects: Protein Secondary Structure Prediction | Gene Expression Exploration | DNA Sequence Classification | Breast Cancer Detection in Women | Protein Thermostability Prediction

Research Interests: AI-driven drug discovery and design | Machine learning applications in pharmaceutical research

Research Papers In Progress: Assessing the Validity of 3D Structural Patches for the Classification of Point Mutations | ML-Based Gait Analysis for TKA Rehabilitation | NLP and ML Approaches to the Endometriosis-Fibromyalgia Link

Volunteering: TEDx Marketing Manager, Hadrumet (2025) | Global Ambassador, IEEE Day Region 8 (2022)